

Question				Marking details				Marks Available					
								AO1	AO2	AO3	Total	Maths	Prac
4 FT	(a)	(i)		141 56 Ba (2) + 92 36 Kr + 3 1 0 n (1)					3		3	3	
		(ii)		Fission , a moderator , neutrons , control rods (1) × 4				4			4		
		(iii)		Uranium isotope	Number of protons	Number of neutrons			2		2	2	
			uranium-235	92	143								
			uranium-236	92 (1)	144 (1)								
	(b)	(i)		3 rd and 4 th statements correct: The time taken for the number of undecayed nuclei to halve ✓(1) The time taken for the activity to halve ✓(1)				2			2		
		(ii)		One half life (1) So ½ (1) ½ alone as answer (2)				1	1		2		
		(iii)		U-235 will have gone through more than 6 half lives / U-235 decays quicker (1) Therefore {fraction/amount} (of U-235) must be smaller so don't agree (1) OR U-238 has only gone through 1 half life /U-238 decays slower (1) Therefore {fraction/amount} of (of U-238) must be larger so don't agree (1)						2	2		
				Question 4 total 15				7	6	2	15	5	